Firm-level price wedges: how good is the mapping, and can it be improved?

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What is being measured

A firm's price wedge is not estimated directly. Wedges are estimated for decile portfolios, and a mapping carries those onto individual firms through their characteristics. The question is how much that mapping can be trusted, and whether it can be improved.

Four criteria, because they answer different questions.

Leaving one characteristic out. Fit on the portfolios of all but one characteristic, predict that one's ten deciles, rotate. Can the mapping price an anomaly it has not seen?

Leaving one family out. The same, holding out a whole Freyberger– Neuhierl–Weber category. Harder and more honest: value characteristics predict each other, so holding out book-to-market while assets-to-market and earnings-to-price remain is not a real test.

Holding out the extreme deciles. Fit on deciles two to nine, predict one and ten. This tests extrapolation, which is what the mapping must do at the firm level: 94% of firm-months sit outside the range the portfolios span on at least one characteristic.

Subsequent returns. Do firms the mapping calls overpriced go on to underperform? This is the only criterion evaluated on firms rather than portfolios, so it is the only one that can penalise bad extrapolation.

Everything is scored on magnitudes, not only ordering: the realised wedge is regressed on the fitted one and the intercept and slope reported. A perfect mapping has intercept zero and slope one. A mapping that ranks firms correctly but compresses the spread shows a slope away from one, and would be no good for a website that reports a wedge rather than a score.

Which portfolios count. Only characteristics whose construction is verified against the replication package enter, as regressors *and* as portfolios. A portfolio we cannot reproduce carries a wedge we cannot trust, so including it adds noise to the left-hand side as well as the right. With 1 characteristic still unverified (aPM), the rule costs nothing: the leave-one-family-out fit of the direct mapping is 0.63 with every characteristic in and 0.62 without it. When six were unverified it raised that fit from 0.56 to 0.67.

Results

Leave one characteristic out

Mapping	R^2	Correlation	Intercept	Slope	RMSE (pp)
3 PCs (paper)	0.587	0.766	-0.19	0.959	6.23
10 PCs	0.747	0.864	-0.12	0.965	4.88
direct	0.766	0.875	-0.11	0.959	4.70
direct, precision-wtd	0.774	0.880	-0.35	0.944	4.63
direct + squares	0.370	0.608	-1.39	0.626	8.48
squares, precision-wtd	0.449	0.670	-1.31	0.640	8.09

Leave one family out

Mapping	R^2	Correlation	Intercept	Slope	RMSE (pp)
3 PCs (paper)	0.389	0.624	-1.68	0.665	8.15
10 PCs	0.467	0.683	-1.70	0.673	7.76
direct	0.619	0.787	-1.01	0.843	6.14
direct, precision-wtd	0.609	0.781	-1.30	0.805	6.33
direct + squares	0.038	0.195	-4.89	0.047	39.54
squares, precision-wtd	0.049	0.222	-4.88	0.056	37.44

Extremes held out

Mapping	R^2	Correlation	Intercept	Slope	RMSE (pp)
3 PCs (paper)	0.737	0.858	+1.02	0.963	7.23
10 PCs	0.805	0.897	+0.04	0.863	6.46
direct	0.804	0.897	-1.35	0.853	6.56
direct, precision-wtd	0.808	0.899	-1.37	0.856	6.50
direct + squares	0.571	0.756	+0.11	0.798	9.51
squares, precision-wtd	0.626	0.791	-1.52	0.699	9.72

Do the wedges predict returns?

Firms are sorted into deciles on their fitted wedge each month and tracked forward. Decile 1 is the most negative wedge (most underpriced), decile 10 the most positive, so if wedges are mispricing that resolves, decile 1 should outperform and the spread should be positive.

Mapping	Horizon	Weighting	Decile 1	Decile 10	Spread (pp)
3 PCs (paper)	12m	cap	18.4	10.7	7.7
3 PCs (paper)	36m	cap	60.3	34.8	25.5

Mapping	Horizon	Weighting	Decile 1	Decile 10	Spread (pp)
3 PCs (paper)	60m	cap	118.3	69.4	48.9
3 PCs (paper)	12m	equal	26.3	11.5	14.8
3 PCs (paper)	36m	equal	80.4	35.1	45.3
3 PCs (paper)	60m	equal	143.2	67.0	76.2
10 PCs	12m	cap	21.3	10.5	10.7
10 PCs	36m	cap	71.1	33.4	37.7
10 PCs	60m	cap	134.3	65.6	68.8
10 PCs	12m	equal	27.2	10.5	16.7
10 PCs	36m	equal	84.1	31.6	52.6
10 PCs	60m	equal	150.0	62.3	87.7
direct	12m	cap	18.9	10.1	8.8
direct	36m	cap	64.6	31.9	32.7
direct	60m	cap	125.1	62.4	62.8
direct	12m	equal	25.5	10.0	15.5
direct	36m	equal	80.4	30.2	50.2
direct	60m	equal	145.9	58.9	87.0

The wedges predict returns strongly and with the right sign. Over five years the direct mapping separates the extreme deciles by 63 percentage points capitalisation-weighted and 87 equal-weighted, and its deciles are monotone except capitalisation-weighted at 36 months. Three principal components produce a smaller spread (49 and 76 points) and are not monotone capitalisation-weighted at 12, 36 and 60 months.

Can the mapping be improved?

Precision weighting: no, and the reason is informative

If some portfolio wedges were estimated far more precisely than others, the regression should weight them accordingly. They are not. Reproducing the package's own standard error — a Newey–West variance on the cohort-level ratios carried through the logarithm by the delta method — and separately a moving-block bootstrap with the package's 180-month blocks:

	Median	10th pct	90th pct	Max
Standard error (pp)	29.0	27.1	32.2	40.6

The ratio of the 90th to the 10th percentile is 1.2. Precision is close to uniform across the 560 portfolios, so weighting by it has almost nothing to exploit — and in the event it is very slightly worse on all three criteria, presumably because the weights add estimation noise of their own.

What the exercise does reveal is the *level*: a standard error around 29 percentage points on wedges whose cross-sectional standard deviation is about 10. Individual portfolio wedges are imprecise, because 463 formation cohorts that overlap in 179 of their 180 months are worth only two or three independent observations. That is a caveat on the whole exercise rather than on any mapping.

Non-linear terms: no

Adding squared ranks roughly halves out-of-sample fit and pushes the slope to 0.6 or below. The design already carries the non-linearity that matters: using all ten deciles rather than the extremes lets the wedge be non-linear in each sorting characteristic, and squaring 56 standardised ranks mostly adds collinear regressors for 560 observations to overfit.

Shrinkage: yes, and it is the substance

Unpenalised, the direct regression fits best in sample and generalises worst. Penalised at the level chosen inside each fold, it is the best mapping on every criterion at once and is close to unbiased in magnitude. The penalty is not a technicality: without it the fitted wedges of unfamiliar portfolios are more than twice too dispersed. On the leave-one-family-out test the realised wedge moves 0.42 for each point of fitted wedge with no penalty, against 0.89 with the penalty of 100 (run of 12 September 2026, `group_cv.held_out` on the verified portfolios).

Number of components

Ten principal components come close to the direct regression and beat three decisively on every criterion. If a dimension-reduced mapping is wanted for interpretability, ten is the defensible choice; three is not.

Recommendation

Use the penalised direct regression of portfolio wedges on portfolio characteristic ranks, fitted on verified characteristics only, with all ten deciles. Report firm-level wedges together with an indication of how far a firm's characteristics sit outside the range the portfolios span, because that is where the magnitudes stop being trustworthy and no portfolio-level criterion can detect it.